THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE MCQ TEST SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. How many non-square numbers lie between 9^2 and 10^2? 1

(a) 18 **(b)** 9 **(c)** 19 **(d)** 17

2. The value of $\dfrac{1}{2^{-3}}$ is: 1

(a) 8 **(b)** 6 **(c)** -8 **(d)** $\frac{1}{8}$

3. Which of the following is NOT a correct Roman numeral? 1

(a) IX **(b)** XL **(c)** IC **(d)** IV

4. The angles of a quadrilateral are in the ratio 1:2:3:4. The largest angle is: 1

(a) 132° **(b)** 144° **(c)** 108° **(d)** 156°

5. Which of the following numbers is divisible by 11? 1

(a) 3464 **(b)** 3465 **(c)** 3466 **(d)** 3467

6. $(x+3)^2$ equals: 1

(a) x^2+6x **(b)** x^2+9 **(c)** x^2+6x+9 **(d)** x^2+3x+9

7. The ratio of 2 m 40 cm to 60 cm is: 1

(a) 1 : 4 **(b)** 4 : 1 **(c)** 4 : 2 **(d)** 5 : 1

8. A value changes from 80 to 100. The percentage increase is: 1

(a) 50% **(b)** 25% **(c)** 22% **(d)** 28%

9. The legs of a right triangle are 5 cm and 7 cm. Its hypotenuse (in cm) is: 1

(a) 12 **(b)** $\sqrt{74}$ **(c)** $2\sqrt{3}$ **(d)** $5\sqrt{3}$

10. If A:B = 2:3 and B:C = 4:5, then A:B:C is: 1

(a) 8:12:15 **(b)** 2:3:5 **(c)** 8:15:8 **(d)** 2:7:5

11. A polyhedron has 12 vertices and 18 edges. The number of faces is: 1

(a) 32 **(b)** 28 **(c)** 6 **(d)** 8

12. The median of 7, 18, 50, 6, 14, 44, 22, 5 is: 1

(a) 14 **(b)** 20.75 **(c)** 18 **(d)** 16

13. Think of a number n. Add 3, multiply by 5, then subtract 15. The result is: 1

(a) $5n$ **(b)** $3n$ **(c)** $n+3$ **(d)** $5n+3$

14. The area of a rectangle is 360 cm² and its length is 24 cm. Its breadth (in cm) is: 1

(a) 180 **(b)** 15 **(c)** 336 **(d)** 17

15. The value of $15^2 - 14^2$ is: 1

(a) 28 **(b)** 30 **(c)** 15 **(d)** 29

16. The value of $(-3)^4$ is: 1

(a) 81 **(b)** 12 **(c)** -81 **(d)** 85

17. The decimal value of 243_5 is: 1

(a) 73 **(b)** 75 **(c)** 71 **(d)** 78

18. In a parallelogram ABCD, $\angle A = (2x + 40)^\circ$ and $\angle C = (3x + 10)^\circ$. The value of x is: 1

(a) 35 **(b)** 29 **(c)** 30 **(d)** 31

19. Which of the following numbers is divisible by 6? 1

(a) 4518 **(b)** 4517 **(c)** 4520 **(d)** 4519

20. $(x - 2)^2$ equals: 1

(a) $x^2 - 4x + 4$ **(b)** $x^2 - 2x + 4$ **(c)** $x^2 + 4x + 4$ **(d)** $x^2 - 4$

21. If A : B = 2 : 3 and B : C = 6 : 5, then A : B : C is: 1

(a) 4 : 6 : 5 **(b)** 2 : 3 : 30 **(c)** 2 : 6 : 5 **(d)** 2 : 3 : 5

22. 18 is what percent of 72? 1

(a) 25% **(b)** 22% **(c)** 28% **(d)** 50%

23. The legs of a right triangle are 7 cm and 9 cm. The area of the square drawn on its hypotenuse (in cm²) is: 1

(a) 16 **(b)** 256 **(c)** 63 **(d)** 130

24. 18 men can do a piece of work in 10 days. 12 men will do it in (days): 1

(a) 15 **(b)** 18 **(c)** 180 **(d)** 11

25. A Koch curve starts with one segment. The number of segments at stage 1 is: 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 5

26. The mean of 5 numbers is 16. If 4 of them are 12, 18, 20, 10, the remaining number is: 1

(a) 20 **(b)** 16 **(c)** 25 **(d)** 15

27. The sum of 3 consecutive integers starting with n is: 1

(a) $n + 3$ **(b)** $3n$ **(c)** $3n + 3$ **(d)** $3n + 4$

28. The area of a triangle is 72 cm² and its base is 16 cm. Its height (in cm) is: 1

(a) 12 **(b)** 11 **(c)** 9 **(d)** 18

29. The triangular number T_8 (sum of 1 to 8) is: 1

(a) 44 **(b)** 64 **(c)** 28 **(d)** 36

30. The standard form of 60000000 is: 1

(a) 6×10^8 **(b)** 6×10^7 **(c)** 60×10^7 **(d)** 6×10^6

31. The number 87 written in base 5 is: 1

(a) 321 **(b)** 223 **(c)** 322 **(d)** 323

MASTERMIND

32. In rhombus ABCD, $\angle ABC = 70^\circ$. The angle $\angle OBC$ (O is the point where the diagonals meet) is: 1

(a) 70° **(b)** 110° **(c)** 35° **(d)** 55°

33. For the number 2538, (largest 4-digit arrangement of its digits) - (smallest arrangement) is: 1

(a) 6174 **(b)** 6264 **(c)** 6183 **(d)** 6084

34. $(x + 3)(x - 3)$ equals: 1

(a) $x^2 + 9$ **(b)** $x^2 - 9$ **(c)** $x^2 - 6x + 9$ **(d)** $x^2 + 6x - 9$

35. On a map, 1 cm represents 5 km. A distance of 3.5 cm on the map is actually: 1

(a) 8.5 km **(b)** 1.75 km **(c)** 22.5 km **(d)** 17.5 km

36. A cycle bought for ₹ 1500 is sold for ₹ 1800. The gain percent is: 1

(a) 20% **(b)** 22% **(c)** 18% **(d)** 40%

37. Taking $m = 8$ in $(2m,\ m^2 - 1,\ m^2 + 1)$, the Pythagorean triplet is: 1

(a) 16, 64, 65 **(b)** 16, 63, 66 **(c)** 16, 63, 65 **(d)** 8, 63, 65

38. On a map the scale is 1 cm = 15 km. Two places 135 km apart are shown on the map at a distance (in cm) of: 1

(a) 2025 **(b)** 9 **(c)** 120 **(d)** 10

39. In the Sierpinski triangle (area of the original triangle taken as 1), the fraction of area still shaded after stage 2 is: 1

(a) $\frac{8}{16}$ **(b)** $\frac{1}{16}$ **(c)** $\frac{6}{8}$ **(d)** $\frac{9}{16}$

40. The mean of 5 observations 39, 11, 19, 36, 15 is 24. If a new observation 12 is added, the new mean is: 1

(a) 22 **(b)** 24 **(c)** 27 **(d)** 36

41. In a number pyramid each block is the sum of the two below it. The bottom row is 1, 2, 3, 4. The top block is: 1

(a) 10 **(b)** 20 **(c)** 22 **(d)** 15

42. In a quadrilateral, a diagonal of 20 cm has perpendiculars of 6 cm and 4 cm drawn to it from the opposite vertices. Its area is: 1

(a) 60 cm² **(b)** 30 cm² **(c)** 100 cm² **(d)** 200 cm²

43. The sum of the consecutive triangular numbers T_6 and T_7 is: 1

(a) 50 **(b)** 49 **(c)** 36 **(d)** 56

44. Which of the following equals 4.07×10^{-3}? 1

(a) 0.0407 **(b)** 0.000407 **(c)** 0.00407 **(d)** 4070

45. A Babylonian numeral has 2 group(s) of 60 and 15 units. Its value is: 1

(a) 27 **(b)** 17 **(c)** 135 **(d)** 105

46. The diagonals of a rectangle ABCD meet at O and $\angle AOB = 80^\circ$. Then $\angle OAB$ is: 1

(a) 50° **(b)** 40° **(c)** 100° **(d)** 80°

47. The sum of the three consecutive numbers 12, 13 and 14 is: 1

(a) 39 **(b)** 37 **(c)** 42 **(d)** 36

MASTERMIND

48. $(x+2)(x+5)$ equals: 1

(a) $x^2+10x+7$ **(b)** x^2+7x+7 **(c)** $x^2+7x+10$ **(d)** x^2+10

49. 8 workers can finish a job in 15 days. 12 workers will finish it in: 1

(a) 12 days **(b)** 10 days **(c)** 11 days **(d)** 22.5 days

50. The marked price of a jacket is ₹ 2500 and it is sold for ₹ 2000. The discount percent is: 1

(a) 18% **(b)** 40% **(c)** 20% **(d)** 22%

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND